



GE Medical Systems

Technical Publication

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Revision 3

Common Platform Q4 2004 Release

Application Tips and Work-Arounds

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Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

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REVISION HISTORY

Revision	Date	Reason for change
1	4/29/2004	Initial Release of document. Supports the 04MW13.7 release. This release supports Xstream Upgrades to H1/H2 based LightSpeed systems , WideView Upgrade to LightSpeed 3.X 4 slice systems.
2	5/24/2004	Update document for additional workarounds and add system effectivity for LightSpeed Pro ¹⁶ 80 and 100 kW, LightSpeed RT and LightSpeed 4,8,16 MDAS Xstream systems.
3	3/21/2005	Update document for 04MW44.9 Release. This release supports Xstream based LightSpeed 1.X, 2.X, 3.X and 4.X systems, LightSpeed 5.X based systems, LightSpeed ¹⁶ , LightSpeed Pro16 and RT systems and Xstream based HiSpeed QXi systems

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INTRODUCTION

This Tips and Workaround document applies to the following products running version 04MW44.9 software. The Legacy precautions previously contained in this document have been moved to the LightSpeed Series Learning and Reference Guide Rev. 5

- Xstream upgrade to H1 and H2 SDAS LightSpeed systems
- **LightSpeed 5.X platform**
 - LightSpeed RT
 - LightSpeed Pro¹⁶
 - 80 kW option
 - 100 kW option
 - LightSpeed¹⁶
 - LightSpeed Ultra
 - LightSpeed Plus
 - LightSpeed QX/i
- **LightSpeed 3.X and 4.X platform (Reference of cradle needed after reboot with flashing 8's displayed on gantry display panel)**
 - LightSpeed QX/i
 - LightSpeed Plus
 - LightSpeed Ultra
 - LightSpeed¹⁶

The document is divided into the following parts

Features unique to LightSpeed 5.X platform

Features common to all platforms

New Precautions for 04MW44.9 software

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NEW FEATURES LIGHTSPEED 5.X PLATFORM

AUTO MA UPDATES

Updates have been made for LightSpeed 5.X based LightSpeed¹⁶ systems to assure more consistency in realized noise in the image based on the Noise Index prescribed. Noise index prescribed and Noise realized (Standard Deviation measured) in the image will now be more closely matched.

LightSpeed 5.X based LightSpeed¹⁶ systems previously running 04MW23.1 software may see a difference in mA required or different noise realized in the image for a given scan due to these changes. For customers who have a LightSpeed 4.X based LightSpeed¹⁶ (Reference of cradle needed after reboot with flashing 8's displayed on gantry display panel).and LightSpeed 5.X based LightSpeed¹⁶ systems there may be slight differences seen in image noise when using the same prescribed noise index.

Please use the following conversion to adjust your protocol to achieve the noise you desire in the image.

For LightSpeed¹⁶ systems running software version 04MW23.1 upgrading to version 04MW44.9.

To obtain the same image noise

For Large SFOV decrease 4 clicks in dose steps. This will raise the prescribed Noise Index 20 % and will result in the same measured Standard Deviation as the previous release. If no change in Noise Index is made then the measured Standard deviation in the image will be lower (less noise).

For Ped Head, Head or Small SFOV increase 2 clicks in dose step. This will lower the prescribed Noise Index 10% and will result in the same measured Standard Deviation as the previous release. If no change is made to the Noise Index the measured Standard Deviation in the image will be higher (more noise).

For LightSpeed 5.X based LightSpeed¹⁶ to match LightSpeed 4.X based LightSpeed¹⁶ systems

To match image noise between the systems on the LightSpeed 5.X based LightSpeed¹⁶ decrease 1 click in Dose Step. This will raise the Noise Index 5%. LightSpeed 5.X based systems need to use a Noise Index 5% higher than the noise Index used on the LightSpeed¹⁶ 4.X system.to obtain equal image noise.

Sample Auto mA user interface screens

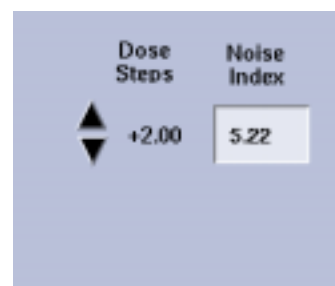
Auto mA User Interface Screen



No change in Dose step



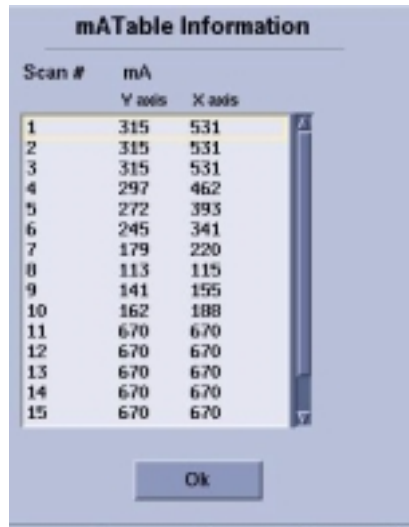
10% change NI 2 clicks in Dose Step



SMART MA

SmartmA provides X-Y modulation of mA. SmartmA is used in conjunction with Auto mA. The use of SmartmA can reduce dose to the patient approximately 15% with no more than a 7% increase in image noise.

SmartmA is enabled in the Auto mA pop up and can be enabled on a group by group basis. When enabled a high and low mA value are calculated for each Z location. The high and low mA values are calculated from the long and short axis ratio of the patient. During scanning the mA will be varied 4 times per rotation when SmartmA is on. When enabled the mA table will show the X and Y mA values.



Scan #	mA	
	Y axis	X axis
1	315	531
2	315	531
3	315	531
4	297	462
5	272	393
6	245	341
7	179	220
8	113	115
9	141	155
10	162	188
11	670	670
12	670	670
13	670	670
14	670	670
15	670	670

To enable SmartmA

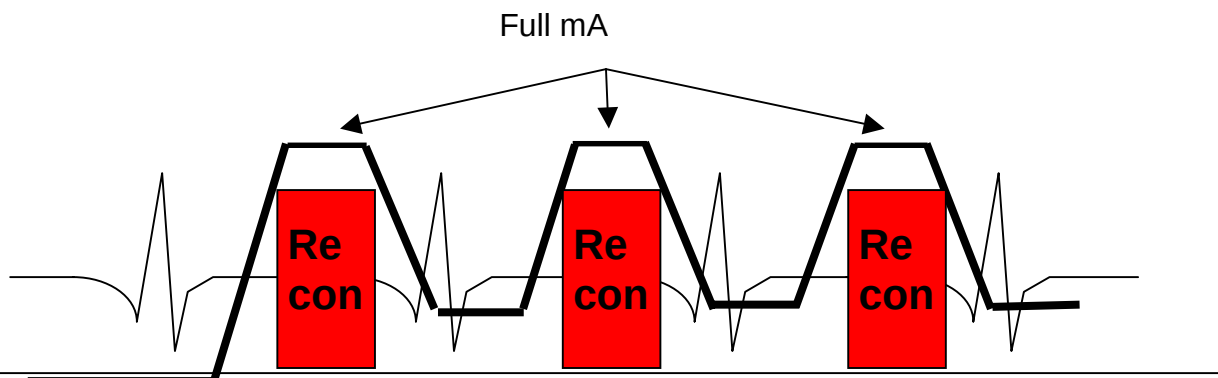
Select mA

Select SmartmA



Please see the Building Protocols chapter in the Learning and Reference Guide for more information.

ECG MODULATED MA (LIGHTSPEED PRO¹⁶ AND 5.X BASED LIGHTSPEED SYSTEMS ONLY)



ECG Modulated mA cont.

- ECG Modulation allows the user to vary (modulate) the mA delivered in relation to the phase % of the R-R interval of the patient's heart cycle.
- In a stable heart rate this can lead to clinically relevant dose reduction for the patient by implementing "full mA" for the useful portion of the cardiac cycle where images are to be reconstructed and a reduced mA for the remainder of the cardiac cycle.
- There are 4 x parameters to set for ECG dose modulation.
 - o Start and End phase for full mA
 - o Min and Max mA Setting
- Set the start phase for full mA to be delivered in the useful portion of the cardiac cycle for imaging (example 65%).
- Set the End Phase for full mA to be delivered in the useful portion of the cardiac cycle for imaging (example 85%).
- Set the Min mA to be used outside the above phase range (no less than 20% of full mA is set, 40% of Max mA is the default).
- Set the Max mA to be used within the above phase range.
- There will be ramp up time factored in for the Min mA to alter to Max mA & vice versa by the s/w .

The image shows a software dialog box titled "mA Control". It has a yellow button labeled "ECG modulated mA" which is currently selected. Below it, there are input fields for "Start Phase" (set to 65) and "End Phase" (set to 85). Further down, there are input fields for "mA Range" with "Min" set to 200 and "Max" set to 500. At the bottom left, there is a blue button labeled "Manual mA" and a text input field containing "500". At the very bottom, there are "OK" and "Cancel" buttons.

Once selected and user returns to the View/Edit screen the mA button displays the annotation as follows to indicate modulated mA with the heart cycle of the patient.

- ECG
- Max mA
- Phase % where maximum mA will be delivered

Images	Scan Type	Start Location	End Location	No. of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time
1-161	Cardiac Burst 0.4 sec.	00.000	1100.000	161	0.625 2.60 0.26:1	0.625	30.0	Large	120	ECG 500 65-85	16.7

INCREASED DAS SAMPLING FOR 0.5 AND 0.4* SECOND SCANS (ONLY AVAILABLE ON GDAS BASED SYSTEMS)

The DAS sampling rate has been increased for 0.5 and 0.4* second rotation times. 0.5 second sample rate is 1968 hz giving 968 views per rotation. 0.4 second sample rate is 2000 hz giving 800 views per rotation. The increased sample rate will provide improved image quality at these rotation times.

*Option on LightSpeed Pro¹⁶ systems

ROTOR MANAGEMENT

Management of the tube rotor has been updated for LightSpeed Pro¹⁶ based systems. The rotor will be allowed to be brought up once every 90 seconds. This time is shortened from every 3 minutes in the previous release. This change extends tube life.

TABLE UP MOTION

When table up motion trips the motion sensor on the gantry, further up motion will be restricted until the interference is cleared.

CARDIAC PITCH

For SnapShot Burst acquisitions using 0.4 second rotation time the pitch has been changed to 0.24:1 to improve image quality.

CARDIAC SMALL SFOV

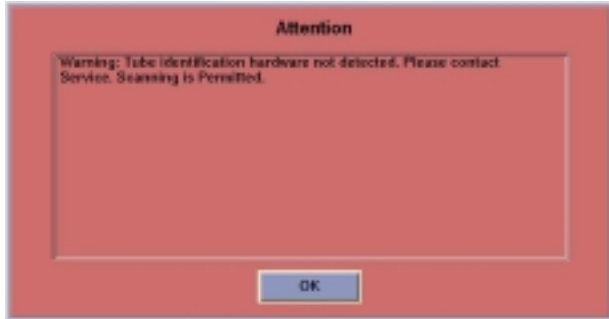
When the scan type is Cardiac Helical and the SFOV is small the dose reported will be based on a 32 cm object size for CTDI and DLP. Prospectively DFOV will be limited to 25 cm, retrospectively the user will be able to recon out to 32cm.

COLOR IMAGE PRINT

Color Saved images may now be filmed to a Post Script printer from the Exam Rx desktop.

SMARTID

SmartID is a hardware and software feature which verifies the correct tube hardware is installed on a system. If the SmartID hardware is not working correctly or is not detected a message will be posted directing the user to contact Service. SmartID is only available on the LightSpeed Pro¹⁶ and LightSpeed RT systems.

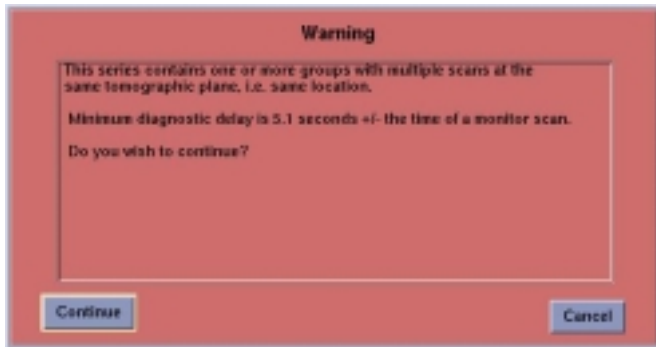


NEW FEATURES COMMON PLATFORM

SMARTPREP

Improvements have been made in Smart Prep to address reliability issues and issues with the transition from Monitor Phase to Scan Phase taking longer than expected.

The diagnostic delay is adjusted at Confirm to account for table movement required between the Monitor and Scan phase locations. If the Diagnostic Delay prescribed is less than what the system can do, the expected delay will be posted at Confirm.



QUICK SNAP

The Recon Status update has been modified to show the range of images and the percentage of reconstruction completed in the Feature Status area.

SOFTWARE VERSION NAMING UPDATE

The software version name reported on the Service Desktop Home page now contains 2 parts software build name and hardware configuration of the system. The format is as follows 04MW44.9.G_T_D_R

$\underbrace{\hspace{1.5cm}}_{\text{build label}} \underbrace{\hspace{1.5cm}}_{\text{Hardware config}}$

Where G equals gantry type, T equals tube type, D equals DAS type and R equals recon type.

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NEW PRECAUTIONS 04MW44.9 RELEASE

SYSTEM

- For LightSpeed 5.X based LightSpeed 16 systems running the table in the short foot print configuration. The restriction in cradle range is not active when the cradle is unlatched. Please take care when moving the cradle to not run into objects or walls.
- When running Quick Snap the Caps lock and Num lock button should be off. If on Quick Snap will fail to complete.
- The clock on the sysem may gain time. Please contact your field engineer to reset the time or if your site has a Network time server the system can be configured to sych time to the timer server when the system is rebooted.

HIPAA LOG IN

- To move between the user name and password field press the tab key twice.

SCAN (ALL SYSTEMS)

- Show localizer does not show the Scan Field of View when prescribing scans. When scanning head studies, verify that you have your patient centered at ISO center to assure that the entire head is included in the SFOV.
- Show localizer may fail to display or may fail to display the slice lines for the scans prescribed. If this occurs turn off show localizer and back on to see if the problem resolves. If the problem does not resolve then turn off Show localizer and display the scout in a free viewport and use Report Cursor to find your scan locations. Manually enter these locations on the view edit screen. There is no need to rescan the patient to obtain the scan locations.
- Click or click and drag with the middle mouse button on the View Edit screen may cause the scanRx process to crash. Avoid using the middle mouse button on the View Edit screen.
- If a scan group contains 1 image and Recon 1 thickness is thicker than Recon 2 thickness. The thinner recon 2 data set may not be contiguous. This occurs due the setting of the start location. If you have multiple groups with 1 image set the start location of Recon 1 at a interval equal to the thinnest recon thickness selected

Scan LightSpeed 5.X Systems

- A scan may fail to resume with an error message unable to bring up Tube Rotor for XXX seconds. The rotor can only be boosted every 90 seconds and the error will occur if the abort shuts off the tube rotor within this 90 second window. Be patient and resume the scan after the time indicated.

PROTOCOL MANAGEMENT

- The Dose information in protocols used from Most Recent does not reflect the dose actually used for the protocol. If the protocol contains manual mA, retype the mA value to update the dose display. If Auto mA is used, the dose will not be reflected correctly in Protocol Management, but will be updated at scan time for the patient being scanned based off a valid scout image.

RECONSTRUCTION

- For 16-slice acquisitions Full recon mode has been disabled for scans with Head or Ped Head SFOV and slice thickness of 0.625 or 1.25mm.
- Anatomy scanned off center from ISO center such as extremities, may have loss of resolution when data is acquired helically. Always make sure to center to anatomy of interest as close to ISO center as possible.
- Images reconstructed from Recon 2 that are thinner than the Recon 1 slice thickness may be reconned with interval equal to the Recon 1 slice thickness. This will occur if the scan groups contain 1 image or if the scans were taken with One More Scan. This may create a non-contiguous data set that if used in Reformat or Volume Viewer may result in less than optimal image quality. To avoid this the start location of Recon 1 recon 2 or Recon 3 must be a interval of the thinnest recon thickness from the previous group.

RECON MANAGEMENT

- Scan data must be saved to a DVD on Linux based systems when Save Scan Data is selected. Make sure that you have a DVD RAM cartridge in the drive to save the data.

DISPLAY (APPLIES TO IMAGE WORKS AND EXAMRX DISPLAY)

- The List Select browser may not launch when List Select is selected on Exam Rx Display or may not be visible on Image Works. This is due to the browser process crashing. A system reboot will be needed to recover.
- The Preset WW and WL settings may be reset to all the same value or may revert back to old values. If this occurs reset the preferences in Exam Rx or Image Works.

VARIVIEWER

Images created in Variviewer Batch RX are flipped R-L with the Left side of the patient on the viewers left. The Right and Left annotation is correct. Flip the image prior to filming or manually flip if the images have been networked to a remote review station.

VOLUME VIEWER (OPTION)

- The system may get extremely slow when saving a 3D or VR model using Save/Recall Save Model. This could cause disruption to scanning. Do not start a model save unless you have 10-20 minutes of free time.

REFORMAT

Reformat may hang if trying to select a camera and more than 20 cameras are installed on the system.

CT PERFUSION 2, 3 OR NEURO ON OC – PURCHASED OPTION

- Do not use data acquired with Expanded CT HU ON on a Advantage Windows Workstation. The results will be lower than expected. Turn the Extended CT HU range to OFF for all studies that will be used for processing CT Perfusion studies.

DENTASCAN

- Do not use Extended CT Number range for images that will be processed in Dentascan. This application does not support Extended CT Number range.

BMD

- Do not use Extended CT Number range for images that will be processed in BMD on an Advantage Windows or Mindways system. These applications do not support Extended CT Number range.

ADD SUBTRACT

- Comb images will not contain the e/s/i used for the added image.

FILMING

- Image annotation may be small or blurry on filmed images when the first image is a scout that is over 600mm. To avoid this place this image on the last space of a film.

NETWORK

- The system no longer supports Advantage Net protocol. When sending images, DICOM send should be utilized.

IMAGE MANAGEMENT

- Removal of images should be done with the system in an idle state. Failing to remove images at idle times can cause image install timeouts that can then cause recon to shutdown and can cause the browser to fail to respond. The user may also see a blank list in the browser. The system should not be shutdown until the image space has been updated fully from the remove. The length of time to synchronize the database is proportional to the number of images being removed. This will assure that all images are reconstructed, displayed and installed into the database without error and that image space is updated correctly.

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